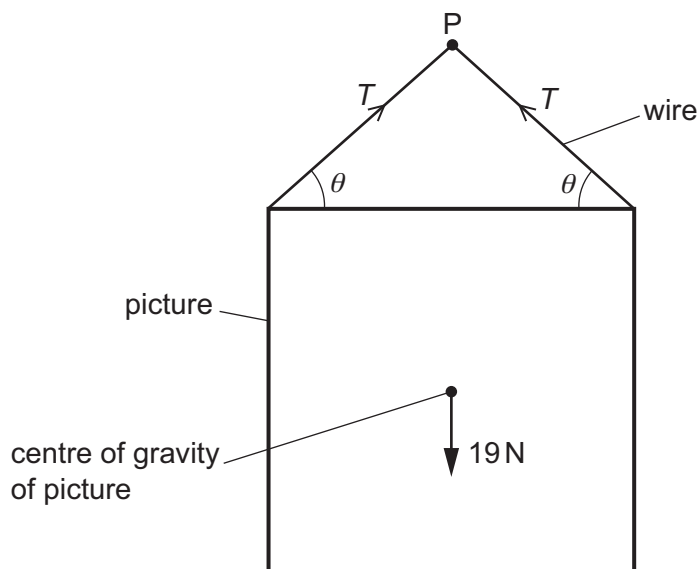


- 4 A uniform square picture is attached to a wire and hangs in equilibrium from a pin at point P, as shown in Figure 4.1.



not to scale

Figure 4.1

The picture has weight 19 N and its centre of gravity is directly below P.

The top side of the picture is horizontal. The wire forms an angle θ with the horizontal. The tension T in the wire is 13 N.

- (a) (i) Define the moment of a force about a point.

.....
 [1]

- (ii) Explain why the moment of the weight about P is zero.

.....
 [1]

- (iii) Calculate θ .

$\theta = \dots\dots\dots^\circ$ [2]

- (iv) The wire has diameter 0.80 mm and is made of a metal with Young modulus 9.3×10^{10} Pa. The wire obeys Hooke's Law.

Calculate the strain ϵ in the wire.

$$\epsilon = \dots\dots\dots [3]$$

- (b) The bottom side of the picture shown in Figure 4.1 is 1.6 m above the ground. The pin at P breaks, and the picture falls vertically downwards. Resistive forces acting on the picture are negligible.

- (i) Calculate the time taken for the bottom side of the picture to hit the ground.

$$\text{time} = \dots\dots\dots \text{ s } [2]$$

- (ii) On Figure 4.2, sketch the variation of the speed of the picture with the distance fallen from its initial position.

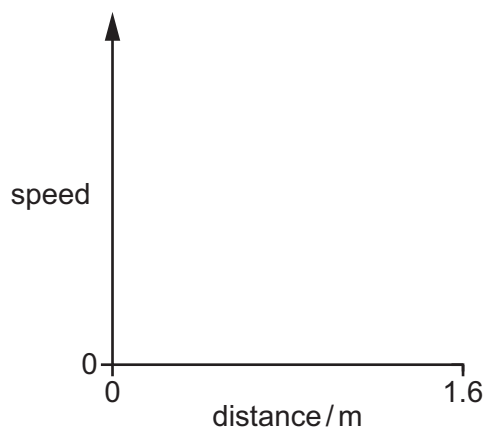


Figure 4.2

[2]

[Total: 11]